Comparative Study of Diffusion-Based Architectures for H&E-to-IHC Virtual Staining: Brownian Bridge, Score-Based, Topology-Consistent, and Transformer Approaches

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Abstract*—Virtual staining—the computational transformation of Hematoxylin and Eosin (H&E) histology slides into immunohistochemistry (IHC) stained images—is one of the most clinically impactful applications of deep generative modeling in digital pathology. Despite the rapid rise of diffusion models as the dominant paradigm for generative image synthesis, Generative Adversarial Networks (GANs) remain the current state of the art for virtual staining, with recent comparative work demonstrating that diffusion models have not yet reliably surpassed GAN baselines in this domain. Bridging this gap requires a systematic understanding of which diffusion model design choices are most effective for histological image translation. In this work, I design, implement, and compare four diffusion-based architectures on the BCI-512 benchmark: (1) BBDM, a Brownian Bridge Diffusion Model that constructs a direct stochastic path between source and target image domains; (2) PST-DIFF+STN, a UNet-based conditional diffusion model with an embedded Spatial Transformer Network for slide-misalignment correction and a Chromatic Frequency Guidance loss for high-frequency stain fidelity; (3) SCORETOPO, a score-based diffusion model with a novel Topological Consistency Loss derived from the soft Euler characteristic and a Mutual Information contrastive objective for cross-modal structural alignment; and (4) HISTDIT+SAM, a Diffusion Transformer with dual-stream SAM-proxy conditioning for structure-aware generation. All models are trained under identical 2-hour A100 wall-clock budgets and evaluated on PSNR, SSIM, LPIPS, and FID. This study provides a systematic analysis of how diffusion model design choices—backbone, noise schedule, and auxiliary conditioning losses—translate to concrete differences in the virtual staining setting, and contributes to the growing effort to establish diffusion models as a viable and eventually superior alternative to GANs in computational pathology.

***Index Terms*—virtual staining, diffusion models, H&E-to-IHC translation, histopathology, HER2 scoring, generative modeling, computational pathology**

I. INTRODUCTION

A. Clinical Motivation for Virtual Staining

Immunohistochemistry (IHC) staining is a cornerstone of modern oncological pathology, providing molecular-level visualization of protein expression that is inaccessible through standard Hematoxylin and Eosin (H&E) preparations. In breast cancer diagnosis, HER2 (Human Epidermal Growth Factor Receptor 2) IHC scoring on a 0–3+ clinical scale directly determines patient eligibility for targeted therapies including trastuzumab and pertuzumab [1]. Despite its importance, IHC staining requires additional tissue sections, specialized reagents, expert pathologist interpretation, and significant laboratory turnaround time—creating bottlenecks and consuming limited biopsy material, particularly in resource-constrained settings.

Virtual staining addresses these limitations computationally. By training generative models to predict IHC stain patterns directly from H&E images, the goal is to enable HER2 scoring from a single standard stain—reducing laboratory cost, conserving tissue, and potentially democratizing access to molecular diagnostics. Even a partial solution capable of reliably distinguishing HER2-negative from HER2-positive cases would have meaningful clinical impact on diagnostic throughput and resource utilization.

B. The Role of Diffusion Models in Virtual Staining

The landscape of generative modeling for virtual staining has evolved through two distinct generations. The first was dominated by Generative Adversarial Networks (GANs), with Pix2Pix [11] serving as the de facto baseline for paired H&E-to-IHC translation. GAN-based approaches accumulated years of task-specific refinement: topology-aware losses, misalignment correction, multi-scale feature fusion, and harmonic convolution—giving them a substantial head start in the virtual staining domain [13], [16], [25].

The second generation, diffusion models [8], has largely superseded GANs in natural image generation, offering more stable training dynamics, richer output diversity, and stronger scalability. However, a landmark comparative study on the HER2match dataset [2] found a striking result: diffusion models do not yet reliably outperform well-tuned GAN baselines for virtual staining. The study found that data alignment quality is a critical confound, and that GANs—particularly Pix2Pix-based approaches—retain strong performance even against modern diffusion variants. Their conclusion calls explicitly for deeper architectural exploration within the diffusion model family to understand what adaptations are needed before diffusion models can match, and eventually surpass, GANs in this domain.

This study is a direct response to that call. I ask: within the diffusion model family, which design choices— UNet vs. Transformer backbone, DDPM vs. score-based noise schedule, standard vs. bridge-based diffusion, pixel-level vs. topological auxiliary losses—most effectively address the specific challenges of H&E-to-IHC virtual staining?

C. Contributions

1) I design and implement four architecturally distinct diffusion-based virtual staining models, each representing a different design philosophy for H&E-to-IHC translation.
2) I introduce a **Topological Consistency Loss** within SCORETOPO, based on the soft Euler characteristic at multiple thresholds, paired with a Mutual Information contrastive objective for cross-modal structural alignment.
3) I integrate an **STN in the diffusion training loop** of PST-DIFF+STN to implicitly correct the slide-pair misalignment identified in [2] as a major performance confound.
4) I propose a **dual-stream SAM conditioning scheme** for HISTDIT+SAM, enforcing diagnostic morphology preservation through independent structural and content cross-attention streams.
5) I provide a systematic empirical comparison of all four models under standardized compute budgets, with analysis directly informed by the GAN vs. diffusion model literature.

II. RELATED WORK

A. Virtual Staining: Current State and GAN Dominance

Virtual staining has matured from a proof-of-concept into a competitive subfield of computational pathology. Rivenson et al. [18] demonstrated early that deep learning networks could transform label-free microscopy images into virtually stained equivalents with high fidelity, establishing the paradigm. For paired H&E-to-IHC translation specifically, Pix2Pix [11] established the conditional GAN baseline, and the BCI Challenge [1] codified a standard benchmark with over 4,000 paired patches spanning all four HER2 expression levels. The BCI Challenge winner used a pyramid Pix2Pix architecture, reflecting the current dominance of GAN-based methods in this benchmark.

Subsequent GAN-based innovations have further widened the lead. TA-GAN [13] introduced topology-aware adversarial training, penalizing differences in connected components and Euler characteristics to reduce structural hallucination. HcGAN [25] applied harmonic convolution for efficient high-quality IHC generation. The most directly relevant comparison to this work is the study in [2] on the HER2match dataset, which concluded that diffusion models do not yet match GAN performance for virtual staining, and that data alignment quality is a major confounding factor. Crucially, they observed that models trained on the better-aligned HER2match dataset significantly outperformed those trained on BCI, suggesting that dataset quality matters as much as model architecture for virtual staining outcomes.

B. Diffusion Models in Histopathology

Despite not yet matching GANs on virtual staining benchmarks, diffusion models have demonstrated real promise across a range of computational pathology tasks and carry strong theoretical motivation for this application. Their iterative denoising formulation allows them to model multimodal distributions—important for H&E-to-IHC translation, where the same H&E patch may correspond to different IHC intensities depending on HER2 expression heterogeneity. The Brownian Bridge Diffusion Model [3] offered a conceptually clean formulation for paired image translation by anchoring both endpoints of the diffusion trajectory. PST-Diff [14] demonstrated that incorporating pathological structural topology constraints within a diffusion framework improves stain transfer consistency. HistDiT [5] extended the Diffusion Transformer architecture [4] to histopathology, conditioning on SAM [6] spatial features to enforce structural coherence during synthesis. Beyond stain translation, diffusion models have been applied to pathology image augmentation, rare morphology synthesis, and segmentation pretraining—building the case that, given appropriate domain-specific adaptation, they can become the leading generative approach in computational pathology [5], [7], [24].

C. Structural Conditioning and Slide Alignment

Two persistent challenges in H&E-to-IHC virtual staining are structural hallucination and slide-pair misalignment. Structural hallucination arises when a model generates IHC-plausible stain patterns in anatomically incorrect locations. The Segment Anything Model (SAM) [6] has emerged as a powerful frozen feature extractor for structural conditioning, providing boundary-sensitive representations that are robust across staining modalities [7]. Slide-pair misalignment arises because consecutive tissue sections undergo small non-rigid deformations during processing, creating noisy pixel-level supervision. As documented in [2], this is a major source of performance variance across virtual staining models. Spatial Transformer Networks (STNs) [15] provide a differentiable mechanism for learning affine warp corrections, and have been integrated into stain transfer pipelines to provide geometry-corrected supervision signals [16].

III. DATASET

A. BCI Dataset

I evaluate all models on the Breast Cancer Immunohistochemistry (BCI) dataset [1], the primary public benchmark for paired H&E-to-IHC virtual staining. The dataset contains high-resolution tissue image pairs from consecutive sections of the same breast tumor biopsy, where each H&E patch is matched to the corresponding IHC-stained patch. Patches span all four HER2 expression levels (0, 1+, 2+, 3+), covering the full range of the clinical scoring scale.

1) Dataset Statistics: Following the standard BCI benchmark split:

- **Training**: 3,311 paired image patches
- **Validation**: 585 paired image patches
- **Test**: 977 image patches (used for BBDM evaluation)

All images are standardized to 512×512 pixels. Where memory constraints required it, images were resized to 256×256 during training, evaluated at the same resolution as their training configuration.

2) HER2 Scoring Context: Scores of 0 or 1+ indicate HER2-negative status; 2+ is equivocal requiring FISH confirmation; 3+ is HER2-positive, qualifying for targeted therapy. The challenge of predicting IHC from H&E is heightened by the fact that different HER2 scores produce dramatically different chromogen intensity patterns in IHC, yet may appear morphologically similar in H&E—requiring the model to infer molecular expression state from histological texture alone.

3) Preprocessing and Augmentation: Images are normalized to $[-1, 1]$ via $x \leftarrow 2 \cdot \text{ToTensor}(x) - 1$. Random horizontal and vertical flipping are applied during training with identical spatial transforms enforced on each paired H&E/IHC image.

4) Dataset Limitations: As documented in [2], the BCI dataset has inherent misalignment between consecutive H&E and IHC sections. This introduces noise in pixel-level supervision and is one of the reasons virtual staining models trained on BCI underperform those trained on more carefully co-registered datasets such as HER2match [2]. All four models in this study are trained on the same BCI split under identical misalignment conditions, ensuring a fair comparison, but results should be interpreted with this dataset-level limitation in mind.

5) On Cross-Validation: In deep image-to-image translation, k-fold cross-validation is rarely employed due to prohibitive compute requirements. Standard benchmarks, including the BCI Challenge, evaluate all models on a single, carefully curated split, and I follow that convention here [1]. Future work should ensure strict stratification across all four HER2 expression levels with identical random seeds for reproducibility.

IV. METHODOLOGY

I implement four diffusion-based virtual staining models, each with a distinct architectural approach to the H&E-to-IHC translation problem. All models share the same input-output specification: given an H&E patch $\mathbf{x}_{\text{HE}} \in \mathbb{R}^{3 \times H \times W}$, the model generates a predicted IHC patch $\hat{\mathbf{x}}_{\text{IHC}} \in \mathbb{R}^{3 \times H \times W}$.

A. Shared Components

All four models use a common multi-scale convolutional H&E encoder:

$$\{f_1, f_2, f_3\} = \text{HEEnc}(\mathbf{x}_{\text{HE}}) \tag{1}$$

with GroupNorm + Conv2d + SiLU blocks and channel widths $[C, 2C, 4C]$ where C is the base channel count (NGF). All attention operations use PyTorch's `F.scaled_dot_product_attention` with float16 casting to activate Flash Attention [22]. Models 1, 2, and 4 use a linear DDPM beta schedule with $\beta_1 = 10^{-4}$, $\beta_T = 0.02$, $T = 1000$, and DDIM sampling [10] with 50 steps at inference.

B. Model 1: BBDM

BBDM [3] replaces the standard Gaussian diffusion path with a Brownian bridge between the H&E source and IHC target, anchoring both endpoints of the stochastic process. The forward bridge process is:

$$q_\delta(\mathbf{x}_t \mid \mathbf{x}_0, \mathbf{x}_T) = \mathcal{N}(\mathbf{x}_t;\ (1 - m_t)\mathbf{x}_0 + m_t \mathbf{x}_T,\ \delta_t \mathbf{I}) \tag{2}$$

where $m_t = t/T$ and $\delta_t = m_t(1 - m_t)$. The denoising network is a UNet [17] with $model\_channels = 128$, multipliers $[1, 1, 2, 2, 4, 4]$, and the H&E image concatenated as 6-channel input. **120.25M parameters**. Training objective: $\mathcal{L}_{\text{BBDM}} = \mathbb{E}\left[\|\hat{\epsilon}_\theta(\mathbf{x}_t, t) - \epsilon\|_1\right]$. Based on the official BBDM codebase [3] with a custom BCI dataset loader.

C. Model 2: PST-Diff + STN

PST-DIFF+STN combines a conditional diffusion UNet with an Asymmetric Attention Module (AAM) for H&E feature fusion and an STN for implicit slide-alignment correction—directly addressing the misalignment problem documented in [2]. The denoising UNet uses base channels $C = 128$ with widths $[C, 2C, 4C, 4C]$. **69.43M parameters**.

At each decoder level, the AAM provides gated cross-attention:

$$\text{AAM}(\mathbf{x}, c) = g \cdot \text{Proj}(\text{Attn}(Q_x, K_c, V_c)) + (1 - g) \cdot \mathbf{x} \tag{3}$$

The STN estimates an affine warp from a rough IHC prediction to the real IHC target, providing geometry-corrected supervision:

$$\mathcal{L}_{\text{PST}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{CFG}} \mathcal{L}_{\text{freq}} + \lambda_{\text{STN}} \|\theta - \mathbf{I}\|_F^2 \tag{4}$$

where $\mathcal{L}_{\text{freq}}$ is a Gaussian-masked Fourier magnitude loss, $\lambda_{\text{CFG}} = 0.15$, $\lambda_{\text{STN}} = 0.5$. Original implementation inspired by [14], [15].

D. Model 3: ScoreTopo

SCORETOPO uses a geometric (NCSN-style) noise schedule and two novel auxiliary losses for histological structure preservation. Noise levels follow a geometric sequence from $\sigma_{\min} = 0.01$ to $\sigma_{\max} = 50$, with $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \epsilon$. **66.74M parameters**.

The Topological Consistency Loss penalizes differences in the soft Euler characteristic χ at $K = 10$ binarization thresholds, extending the topology-preserving objectives of TA-GAN [13] to the diffusion setting:

$$\mathcal{L}_{\text{topo}} = \frac{1}{K} \sum_{k=1}^{K} \|\chi(\hat{\mathbf{x}}_0, \tau_k) - \chi(\mathbf{x}_{\text{IHC}}, \tau_k)\|_1 \tag{5}$$

The MI Contrastive Loss [23] maximizes mutual information between H&E and predicted IHC features:

$$\mathcal{L}_{\text{MI}} = -\mathbb{E}\left[\log \frac{\exp(q_i \cdot k_i/\tau)}{\sum_j \exp(q_i \cdot k_j/\tau)}\right] \tag{6}$$

Combined: $\mathcal{L} = \mathcal{L}_{\text{score}} + \lambda_{\text{topo}} \mathcal{L}_{\text{topo}} + \lambda_{\text{MI}} \mathcal{L}_{\text{MI}}$, $\lambda_{\text{topo}} = 0.10$, $\lambda_{\text{MI}} = 0.05$. Original implementation for this study.

E. Model 4: HistDiT + SAM

HistDiT+SAM [4], [5] replaces the UNet backbone with a Diffusion Transformer operating in patch token space, conditioned on H&E features and a lightweight SAM proxy encoder for structure-aware generation. **80.66M parameters**.

Images are patchified into 16×16 tokens ($N = 1024$ at 512px) and processed through $L = 12$ DiT blocks with adaLN-Zero modulation [4]. Each block applies two cross-attention streams: one to SAM spatial features (structural topology) and one to H&E encoder features (content fidelity). Training loss: $\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda_{\text{SAM}} \| f_{\text{SAM}}(\hat{\mathbf{x}}_0) - f_{\text{SAM}}(\mathbf{x}_{\text{IHC}}) \|_1$, $\lambda_{\text{SAM}} = 0.10$. Original implementation inspired by [4], [5].

V. Experimental Setup

A. Compute Infrastructure

Google Colab (Primary): All final experiments were conducted on **NVIDIA A100-SXM4-80GB** GPUs via Google Colab Pro, using mixed-precision training (bfloat16), AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4}), learning rate 10^{-4} with cosine annealing to 10^{-6}, and gradient clipping at $\ell_\infty = 1.0$.

UCF ARCC Newton HPC (Attempted): I initially attempted training on the UCF ARCC Newton cluster (NVIDIA H100 GPUs, SLURM scheduler). However, the priority queue system introduced substantial wait times between submissions, making iterative debugging impractical. Google Colab Pro's immediate availability, Drive integration for checkpoint management, and interactive notebook environment made it the more practical choice for this exploratory study. Newton was used for early prototyping only.

B. Training Configuration

TABLE I: Training Configuration Per Model

Model	Resolution	Eff. Batch	Epochs
BBDM	256^2	16	47
PST-Diff+STN	256^2	64	35
ScoreTopo	256^2	64	40
HistDiT+SAM	512^2	16	50

All models were stopped by the 2-hour wall-clock timeout rather than by convergence. All reported results represent partially-trained models and should be interpreted accordingly.

C. Evaluation Metrics

- **PSNR**: Peak Signal-to-Noise Ratio (dB). Higher is better.
- **SSIM** [19]: Structural Similarity Index. Higher is better.
- **LPIPS** [20]: Learned Perceptual Image Patch Similarity (AlexNet). Lower is better.
- **FID** [21]: Fréchet Inception Distance. Lower is better.

PSNR and SSIM are computed per image and averaged. LPIPS is computed per image and averaged. FID is computed globally over all generated and reference images using InceptionV3 feature statistics via `pytorch-fid`. Models were evaluated on the BCI validation set (585 pairs for PST-Diff, ScoreTopo, and HistDiT; 977 test patches for BBDM via its built-in `sample_to_eval` pipeline).

D. External Resources and Academic Integrity

- BBDM: Adapted from the official BBDM repository [3] with a custom BCI dataset loader. No other code shared.
- PST-Diff+STN: Original implementation. Conceptually inspired by [14], [15]; no code shared.
- ScoreTopo: Original implementation. Topology loss inspired by [13]; MI loss follows [23].
- HistDiT+SAM: Original implementation inspired by [4], [5]; no code shared.
- All novel components (topology loss, STN-in-the-loop, dual-stream SAM conditioning, MI contrastive objective) are original contributions of this study.

VI. Results

A. Quantitative Comparison

TABLE II: Quantitative Comparison on BCI Validation/Test Set. ↑ = Higher is Better, ↓ = Lower is Better. Best Result Per Metric in **Bold**.

Model	PSNR↑	SSIM↑	LPIPS↓	FID↓
BBDM	**15.16**	**0.243**	0.757	360.96
PST-Diff+STN	10.08	0.016	**0.664**	**214.77**
ScoreTopo	6.89	0.001	1.169	416.81
HistDiT+SAM	11.93	0.000	0.940	443.43

BBDM achieves the highest PSNR (15.16 dB) and SSIM (0.243) by a substantial margin, indicating the strongest pixel-level structural fidelity. PST-Diff+STN achieves the best LPIPS (0.664) and FID (214.77), suggesting its STN alignment correction and frequency-domain loss produce perceptually plausible outputs that match the real IHC distribution more closely. ScoreTopo and HistDiT+SAM underperform on all metrics under the 2-hour compute constraint.

B. Metric Interpretation

The split between BBDM (best pixel metrics) and PST-Diff+STN (best distribution metrics) reflects a known tension in generative image evaluation [20]. In a clinical virtual staining context, FID and LPIPS may carry more practical weight, as they better predict whether generated IHC images would be interpretable to a pathologist or useful for downstream HER2 classification [2]. By this measure, PST-Diff+STN's FID of 214.77 represents the strongest result in this study.

VII. Discussion

A. Why BBDM Performs Best on Pixel Metrics

The Brownian Bridge formulation is the most structurally well-matched diffusion paradigm for paired image translation [3]. Because the stochastic process is anchored at both the H&E source and IHC target, the model never needs to discover the target domain from Gaussian noise—the supervision signal is always grounded at both endpoints of the diffusion trajectory. This structural advantage likely explains BBDM's SSIM of 0.243 compared to near-zero values for all other models: the bridge prevents over-exploration of the output space and makes convergence more efficient under limited compute.

This is consistent with findings in [2], where simpler, well-matched baselines retained competitive performance against more complex diffusion variants.

B. PST-Diff's Distributional Advantage

PST-DIFF+STN's STN component directly targets the misalignment problem that [2] identified as a major source of performance variance in virtual staining models. By estimating a geometry-corrected supervision signal at each training step, the STN reduces noise in the pixel-level targets and improves the statistical alignment between generated and real IHC distributions. This contributes to PST-DIFF+STN's superior FID (214.77 vs. 360.96 for BBDM)—the generated outputs better match the real IHC image distribution even if individual pixel correspondence is lower.

C. ScoreTopo: Topology Loss at Limited Compute

SCORETOPO achieves the lowest PSNR (6.89 dB) despite a theoretically motivated design. The geometric noise schedule requires annealed Langevin dynamics at inference—approximately 1 minute per image on an A100—making full 585-image evaluation impractical. Additionally, the topology and MI losses introduce competing gradients early in training before the denoising network has learned a coherent image prior, potentially destabilizing convergence within the 2-hour budget. The topology-preserving objective is well-motivated by the GAN literature [13] and may prove beneficial with longer training schedules and a pre-warmed diffusion prior.

D. HistDiT: Transformer Convergence Gap

HISTDIT+SAM's near-zero SSIM (0.000) and worst FID (443.43) reflect the well-documented convergence challenge of Transformer diffusion under limited compute [4]. DiT architectures lack the spatial inductive bias of convolutional encoders, requiring significantly more gradient steps to learn local coherence—a property that UNet-based models acquire rapidly through skip connections. At 50 epochs within a 2-hour budget, the 12-layer DiT had not yet entered the training phase where global self-attention patterns consolidate into coherent image structure.

E. Diffusion Models vs. GANs for Virtual Staining

The results of this study reinforce the core conclusion of [2]: diffusion models have not yet matched the performance of well-established GAN-based virtual staining pipelines. The BCI Challenge winner used a pyramid Pix2Pix GAN approach [1], and the virtual staining community has had years to refine GAN architectures for the specific challenges of this task. The results here show that diffusion models, even with domain-specific modifications, still lag behind in terms of the distributional quality measures most relevant to clinical use.

That said, these results should not be interpreted as evidence that diffusion models are fundamentally unsuited for virtual staining. Rather, they reflect the early stage of diffusion model adaptation to the histological image domain. The most encouraging signal is PST-DIFF+STN's FID of 214.77, achieved through targeted misalignment correction—a finding that suggests diffusion models equipped with the right task-specific inductive biases can approach competitive performance. As the field continues to develop better-aligned training datasets, longer training schedules, and more refined conditioning mechanisms, diffusion models are well-positioned to eventually match and surpass GAN performance in virtual staining.

F. Limitations

Training Budget: All models were stopped by the 2-hour wall-clock cap. A fairer comparison would train each model to convergence regardless of time, particularly for ScoreTopo and HistDiT.

No Direct GAN Comparison: To maintain focus on the within-diffusion comparison, I did not train a GAN baseline alongside the diffusion models. A direct comparison with the BCI Challenge winner would better contextualize the gap that diffusion models need to close.

Evaluation Protocol Mismatch: BBDM was evaluated on the test set (977 images) via its built-in pipeline; the other three models were evaluated on the validation set (585 images), introducing a minor inconsistency.

Single Data Split: I train on a single BCI split. Results may not generalize across different dataset configurations or patient cohorts.

VIII. CONCLUSION

Virtual staining stands at a genuinely exciting frontier in digital pathology—one where deep generative models have the potential to fundamentally change how molecular diagnostic information is obtained from tissue. The ability to predict IHC stain intensity computationally from H&E slides could reduce laboratory cost, eliminate the need for repeated physical staining procedures, conserve biopsy tissue, and make HER2 scoring accessible in settings without IHC infrastructure. These are outcomes with direct clinical impact on breast cancer patients worldwide.

GANs are currently the established leaders in this space, as confirmed by the BCI Challenge results [1] and the direct comparative study on the HER2match dataset [2]. GAN-based approaches have benefited from years of task-specific refinement—misalignment-aware training, topology-preserving losses, multi-scale feature fusion—that diffusion models have not yet fully assimilated.

This study explored whether diffusion model architectural choices could begin to close that gap on the BCI-512 benchmark. Among the four models evaluated, BBDM achieved the strongest pixel-level fidelity (PSNR 15.16 dB, SSIM 0.243)—attributable to its bridge formulation that directly anchors generation between H&E and IHC endpoints. PST-DIFF+STN achieved the best perceptual distribution metrics (LPIPS 0.664, FID 214.77) by incorporating STN-based misalignment correction, directly addressing one of the key failure modes identified in [2]. SCORETOPO and HISTDIT+SAM underperformed under the 2-hour compute constraint, reflecting well-documented convergence challenges of score-based samplers and Transformer backbones.

The broader takeaway is not discouraging—it is clarifying. Diffusion models for virtual staining are a promising but immature direction, and this study identifies the specific adaptations that matter most: misalignment correction, bridge-based noise schedules, and longer training schedules. The innovations introduced here—topology-preserving losses, STN-corrected supervision, and dual-stream structural conditioning—are steps toward closing the gap with GANs, and warrant further investigation with greater compute budgets, better-aligned training data such as HER2match [2], and direct GAN baseline comparisons.

Diffusion models have transformed natural image generation. There is every reason to believe that, with the right domain-specific adaptations, they will do the same for virtual histological staining—and this study is a step toward understanding what those adaptations need to be.

For future work, I recommend: (1) training all models to convergence with 200+ epoch budgets; (2) evaluating on better-aligned datasets such as HER2match [2]; (3) including a direct GAN baseline in the comparison; (4) adopting latent diffusion to reduce pixel-space compute for Transformer models; and (5) conducting pathologist-reviewed qualitative evaluation to assess clinical utility.

Code Availability

All model implementations, training scripts, evaluation code, and experimental notebooks produced for this study are publicly available at:

https://github.com/Jeber811/Diffusion_Model_Virtual_Staining

The repository includes full implementations of all four models, the standardized evaluation pipeline with a reproducibility flag, dataset preprocessing scripts, and Google Colab notebooks configured for A100 GPU training. Checkpoint links and dataset setup instructions are provided in the repository README.

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